



Using Generative AI to Adapt Classroom Resources to Meet Diverse Needs

Practical Strategies and Examples



Introduction

Generative AI can **automate and personalize** the adaptation of lessons and materials to meet diverse learner needs.

Research shows that AI tools can transform complex text into simpler formats, break tasks into steps, and generate multimodal content to help all learners access the curriculum.



AI Awareness

I know AI exists and I can use it to do some stuff, including:

- fixing my emails
- making a lesson plan
- generating some quiz questions

Narrow, tool-based “up-skilling”

AI Literacy

My understanding goes beyond technical know-how to encompass human-centered values, ethical considerations, and the ability to critically evaluate and discuss AI concepts, applications, and implications. This includes:

- critically evaluating AI design, output, interests, and implications
- judiciously integrating AI into pedagogy to enhance teaching and learning
- supporting development of students' AI literacy
- developing an awareness and prioritization of data and privacy controls
- foregrounding professional judgement and human agency
- addressing AI in professional learning through a critical lens
- engaging colleagues, students, families, and other stakeholders in meaningful dialogue about AI in teaching and learning.

Professional & philosophical inquiry

AI can cut both ways...

Jess, M., Mitchell, J., Scheuhammer, J., & Spence, R. (2024, April 15). "Framework for accessible and equitable artificial intelligence (AI) in education". *Inclusive Design Research Centre*, OCAD University.
<https://idrc.ocadu.ca/projects/ai-in-education/>

“AI can make things better. It can benefit us all, it can **address inequities**, and it can lower barriers for people with disabilities in education.

It can equally be used to **amplify inequities** (intentional and unintended), including discrimination against people who do not fit a ‘norm.’”

AI Possibilities for Teachers | UDL

Question	Description	Example Generative AI Prompt
Accessible	Who can participate in the lesson and who can not?	Please examine this lesson plan and tell me what type of student would be unable to fully participate in this lesson.
Flexible	Student choice in how they learn and how they demonstrate learning.	Please provide multiple methods for students to demonstrate [learning target/objective].
Free of Bias	What in my learning activity is inadvertently disadvantageous to students?	What components of this lesson assume similar prior knowledge to me, the instructor, or what components are...
Valid	Does my assessment evaluate the specific learning objective I am attempting to assess?	Please change the reading level of this question to a 7th grade level (choose a level that is accessible to all students)

AI Possibilities for Students | UDL

Multiple Means of Engagement	Multiple Means of Representation	Multiple Means of Action and Expression
Automated Summarization Tools	Personalized Learning Materials	Interactive AI Tutors
Gamified Learning Experiences	Language Translation and Support	Adaptive Assessments
Dialogue Simulations for Social Learning	AI-Generated Study Guides and Organizers	Creative Expression Tools

EIPS Approved AI Platforms and Tools

The examples listed below can be run using a variety of AI tools.

The tools listed here are approved for **staff use**. For further details please see the description on the [Artificial Intelligence resource page](#).

Adobe Firefly

Brisk

ChatGPT

Google Gemini

Google NotebookLM

MagicSchool

Microsoft Copilot

Teacher Time Machine



Working Iteratively with AI

- Generative AI tools don't always get it right on the first try. These systems work best when users **engage in iterative dialogue**, gradually refining prompts based on results. This process mirrors how we support student learning: we assess, adjust, and scaffold.
- Asking **follow-up questions** is essential for shaping output that meets diverse learner needs.
- The quality of the result depends heavily on the **clarity and specificity** of back-and-forth interactions between the human and the model.



Providing Context to AI

To generate useful responses, AI tools must understand the task—this requires *context*.

- **Upload Documents:** AI Tools often allow you to upload or link documents, websites, and other resources. The AI can then generate summaries, adapt content, or suggest modifications based on the actual material.
- **Paste Text Directly:** paste the specific text and then give instructions. Starting with concrete input leads to more tailored output.
- **Provide Background Details:** Describe the situation. For example: “I have a student with dyslexia who needs help understanding this Grade 9 social studies passage”. These cues guide the AI to produce output aligned with learner needs.

Context boosts AI accuracy and relevance and it models for students how to be clear and precise when seeking help—an important digital literacy skill.



Examples and Demonstrations

The following slides describe key categories of AI support, why each is helpful for particular exceptionalities, and practical guidance (with example prompts) for teachers and support staff.

- ✓ Text simplification and Language Support
- ✓ Visual and Multi-Modal Supports
- ✓ Executive Function and Organizational Support
- ✓ Content Reformatting and Scaffolding
- ✓ Summarization and Comprehension Checks
- ✓ Social and Behavioural Supports
- ✓ Writing and Communication Assistance



Text Simplification and Language Support

What AI can do: AI chatbots can **paraphrase or summarize complex text**, adjust vocabulary, provide definitions, translate or restate content in the student's first language, or offer dyslexia-friendly fonts and formatting suggestions.

Why it helps: Students with language-based learning disabilities (e.g. dyslexia, reading disabilities) or cognitive delays often **struggle with dense text and long sentences**.

Simplified text reduces processing load and improves comprehension.



Text Simplification and Language Support

How to use it:

- ✓ Use ChatGPT, Microsoft Copilot or Google Gemini with prompts that specify simplification.
- ✓ Brisk Teaching's browser extension can simplify text on any website, and MagicSchool's AI tools can generate leveled texts.
- ✓ In Google NotebookLM ask the assistant to **summarize or outline** a long document.

Remind the AI to use clear, non-technical words and to highlight key ideas. For formatting, prompt the AI to “use 14pt Arial font, left-aligned, dark text” which research shows improves readability for dyslexic readers.



Text Simplification and Language Support

Examples/Prompts:

- “Simplify this Shakespeare sonnet into modern English. Make each line into a plain-language sentence.”
- “Turn this biology article into 3 bullet points, defining any advanced terms.”
- “Rewrite this science paragraph for Grade 7 reading level, using short sentences and bullet points:” (paste original text)
- “Summarize the following history chapter in 5 simple sentences:”



Visual and Multi-Modal Supports

What AI can do: Generative AI can **create custom images, diagrams, charts, infographics, or comic strips** to illustrate concepts. Teachers can also have AI produce *visual schedules*, pictorial storyboards, or symbols for vocabulary.

Why it helps: Visual learners, students with autism, and many others benefit from seeing ideas in pictures. Images and visual organisers support understanding and memory. UDL research highlights that providing information in multiple formats **enhances equity**.

Custom visuals also help students with language delays or ELL.



Visual and Multi-Modal Supports

How to use it:

- ✓ Use Adobe Firefly to generate illustrations or icons.
- ✓ MagicSchool's AI can create quick infographics.
- ✓ Use ChatGPT to generate images based on a social script.

Using AI to produce multimodal content aligns with UDL's emphasis on multiple representation methods. Custom visual aids (stickers, icons, diagrams) have been shown especially useful for students with visual learning needs or communication difficulties.



Visual and Multi-Modal Supports

Examples/Prompts:

- “Create a colorful cartoon diagram of the water cycle with labels (evaporation, condensation, precipitation, collection).”
- “List the classroom morning routine with a simple icon for each step.”
- “Describe a simple comic-strip storyboard about a student asking a question in class and then generate images based on that script.”
- Generate an image of 5 apples and write ‘+5’ and ‘-3’ on it, showing the sum and difference.”
- Write a 3-panel social story (with simple captions) about lining up quietly.” (Then use image AI to illustrate each panel.)



Executive Function and Multimodal Support

What AI can do: AI can **break down complex tasks** into sequential steps, create checklists and schedules, and suggest time-management plans. AI can also answer “What’s next?” questions to guide students through a process.

Why it helps: Students with ADHD, autism, or cognitive disabilities often struggle with planning, organization and prioritizing tasks. Explicit step-by-step guidance and schedules **reduce anxiety and increase independence.**

Allowing a student to ask an AI assistant “What’s my next step?” can serve as on-demand coaching.

Executive Function and Multimodal Support

How to use it:

- ✓ Use ChatGPT, Copilot, Gemini or MagicSchool to create lists and schedules
- ✓ Google NotebookLM (when integrated with your notes) can also generate a timeline or outline from a long to-do list.
- ✓ Brisk and Copilot can suggest task breakdowns in-app.

AI task-analysis tools can automate planning steps.

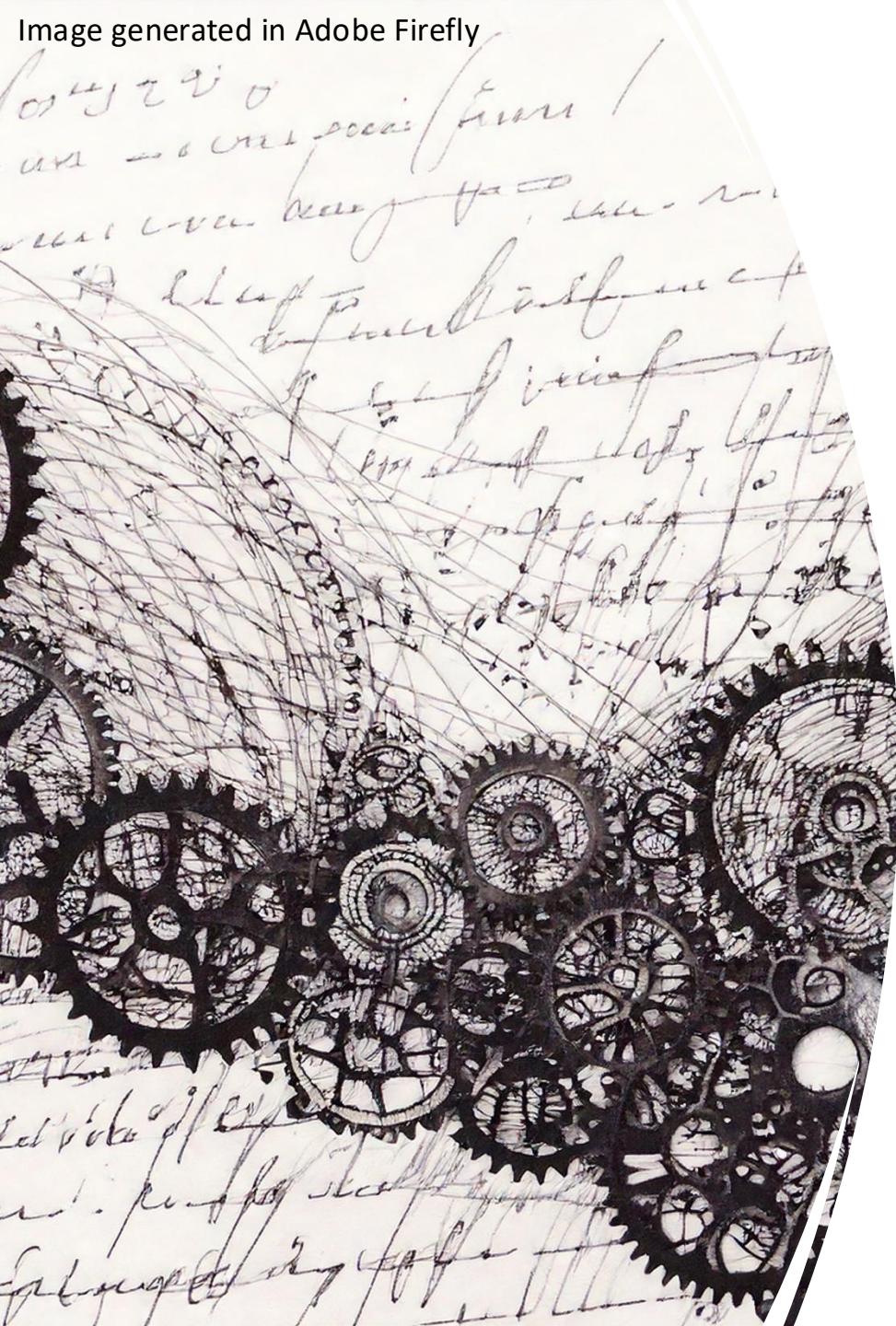
Generative models have also been shown to break projects into parts, which benefits students with executive function challenges. Always review the AI's steps to ensure accuracy and appropriateness.

Executive Function and Multimodal Support

Examples/Prompts:

- “List the step-by-step procedure to solve this chemistry lab experiment.”
- “Create a weekly study schedule for a student who has 2 hours after school on weekdays.”
- “Break down the task of ‘writing a lab report’ into an ordered checklist with deadlines.”
- “Suggest a study plan for this week to review biology and math, including short breaks.”

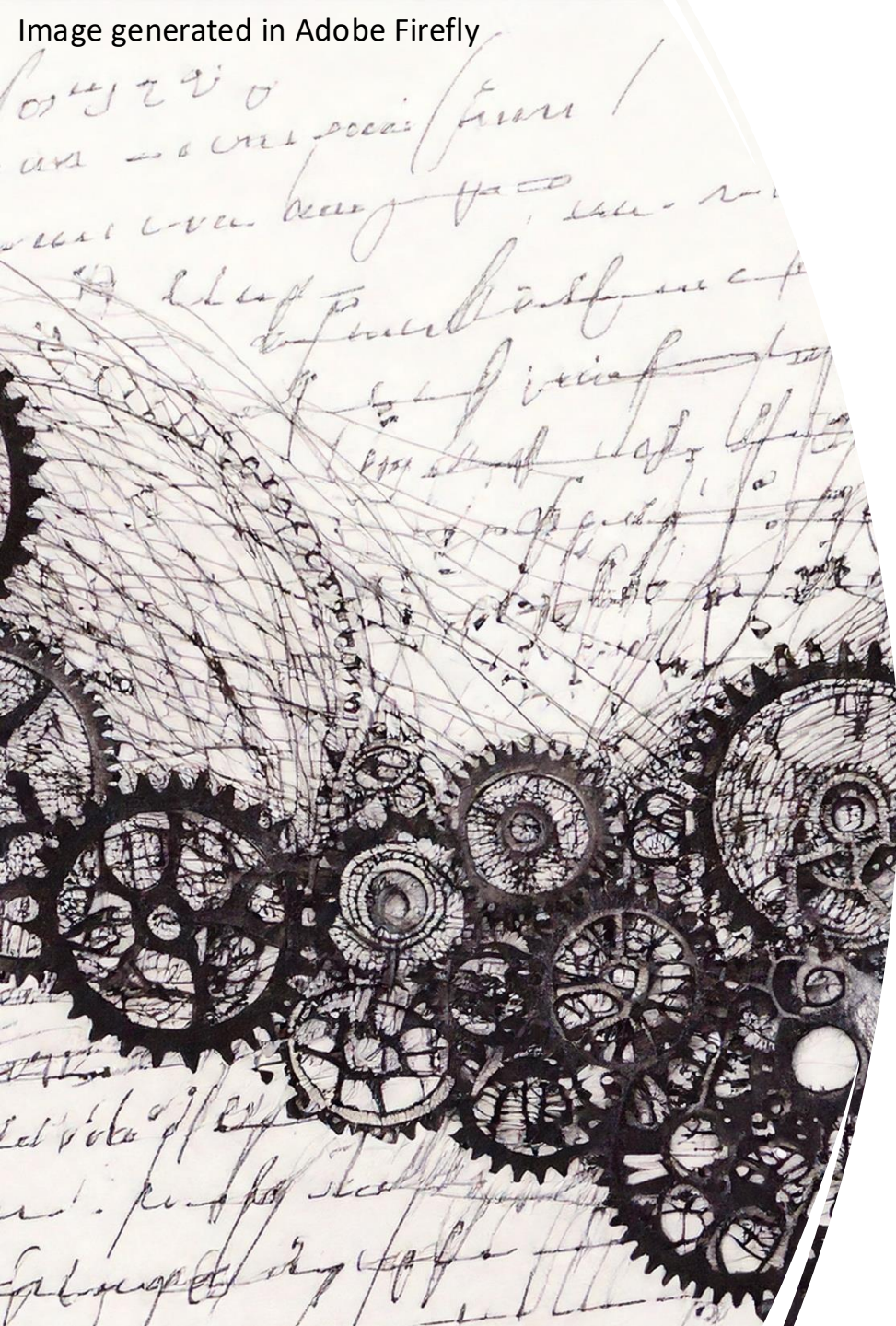




Content Reformatting and Scaffolding

What AI can do: AI can **reformat existing content** into alternative structures or activities. It can also act as a “tutor” by explaining concepts in new ways, giving examples, or providing analogies.

Why it helps: Different students need different entry points to the same content. A lesson reformatted as a diagram, quiz, or story can engage learners who might otherwise disengage. For students with dyslexia or ADHD, interactive formats and guided notes help focus attention. Scaffolding reduces cognitive load. By generating varied representations (text, questions, visuals, outlines), AI creates **multiple means of engagement**, a core UDL principle.

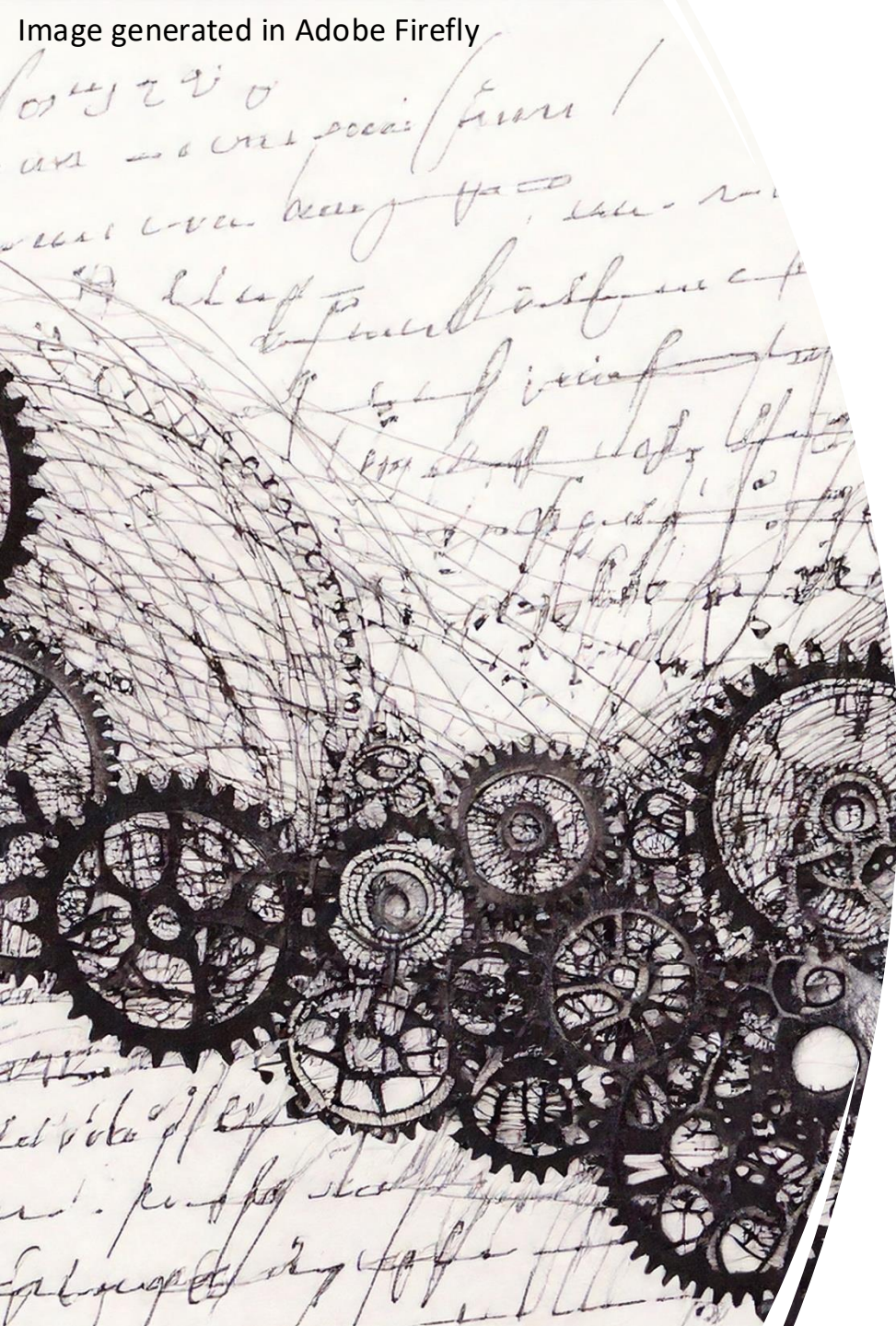


Content Reformatting and Scaffolding

How to use it:

- ✓ Use ChatGPT, Copilot, or Gemini to convert text into other forms.
- ✓ For worksheets, MagicSchool and Brisk can produce quizzes from your content.
- ✓ Google Gemini or NotebookLM can turn lecture notes into flashcards.
- ✓ Teacher Time Machine can import old exam questions and update them.

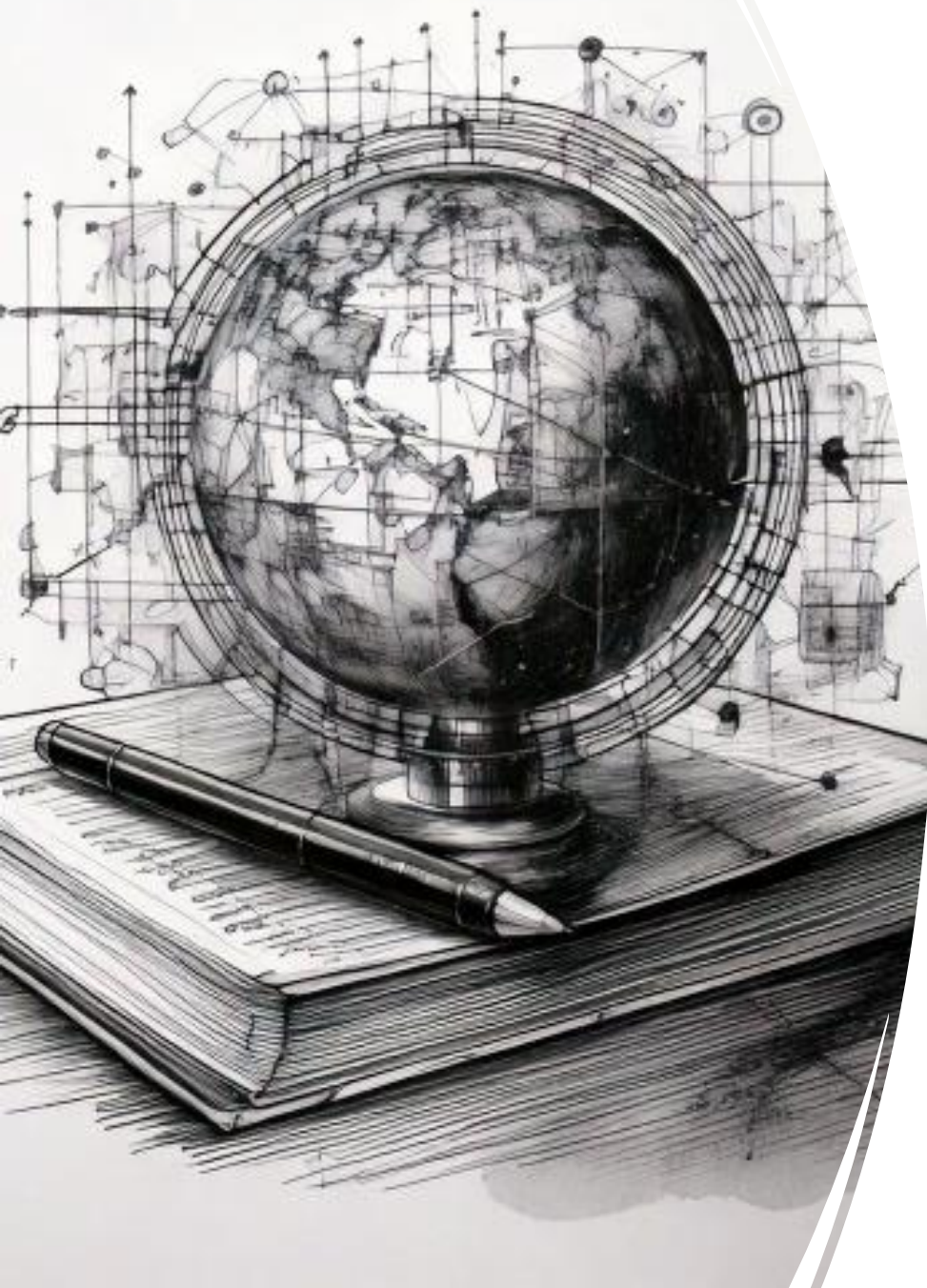
Converting text into simpler formats is an established accommodation. Generative AI can automate this process



Content Reformatting and Scaffolding

Examples/Prompts:

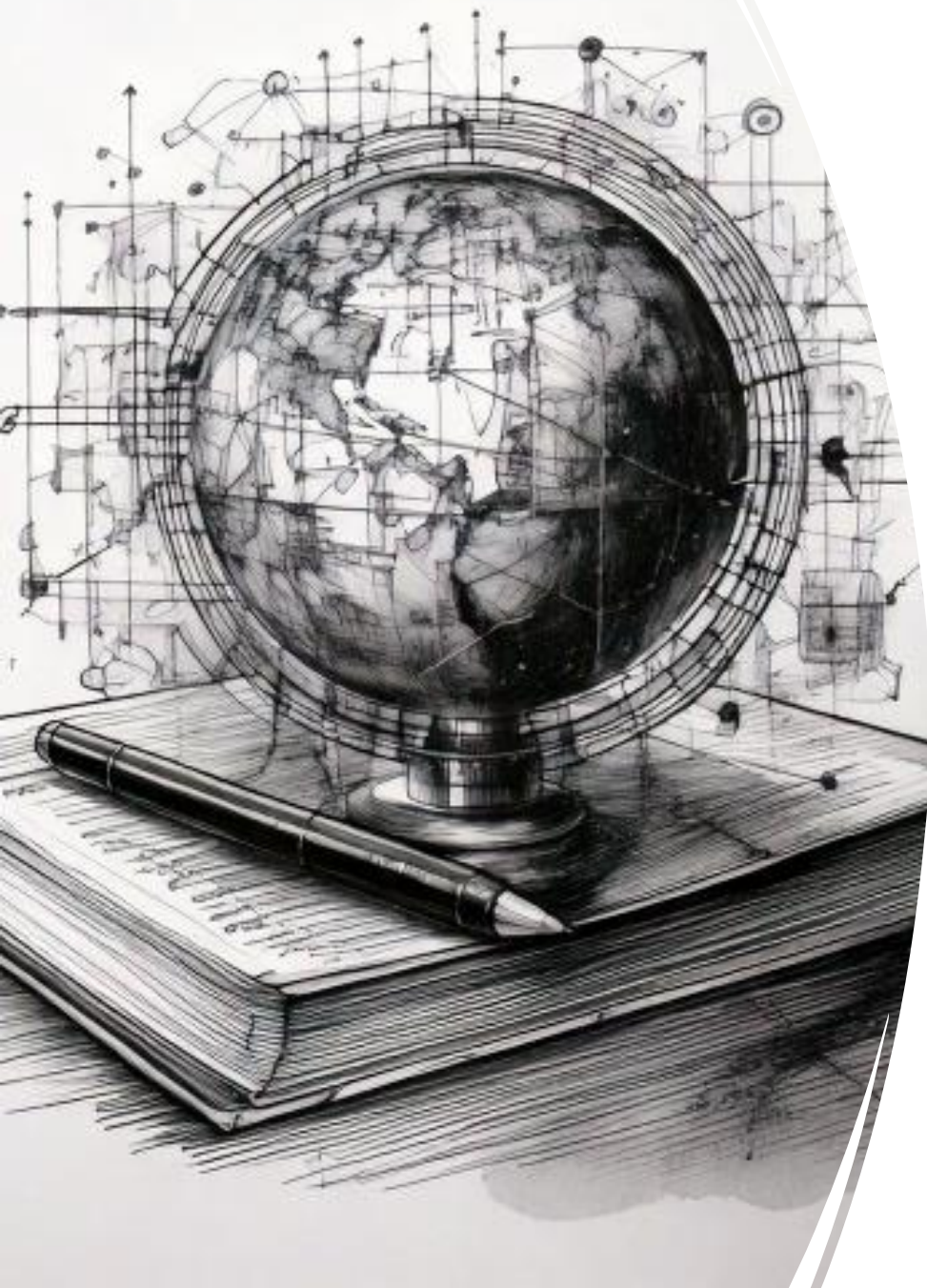
- “Create 5 true/false questions and answers about this social studies paragraph.”
- “Rewrite this poem into a simple rhyme for kids.”
- “Explain this concept as an emoji story,”
- “Give a real-life example of this math concept.”
- “List 5 fill-in-the-blank sentences to practice the word ‘photosynthesis’.”
- “Turn this algebra problem into a step-by-step worked example (with each step on a new line).”



Summarization and Comprehension Checks

What AI can do: AI can **summarize long texts or lectures**, generate comprehension questions, condense chapters into key points, or answer specific questions. They can also explain why an answer is correct/incorrect.

Why it helps: Students who struggle with working memory or attention (e.g. ADHD, learning disabilities) benefit from having the **“big picture” explicitly stated**. Comprehension questions help monitor understanding and review.

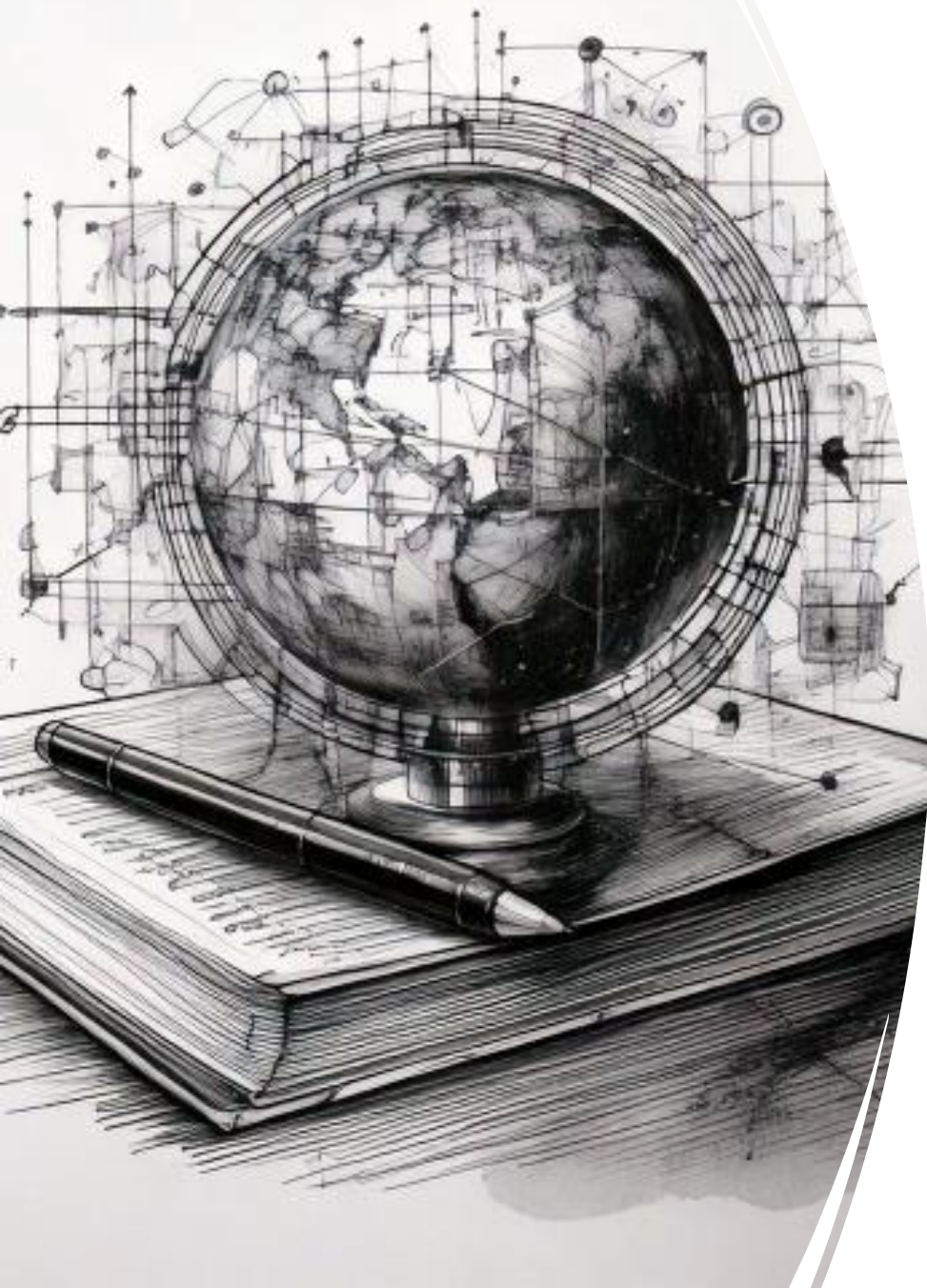


Summarization and Comprehension Checks

How to use it:

- ✓ Prompt ChatGPT, NotebookLM, or other chatbots to summarize the main ideas of a text in 4 bullet points.
- ✓ Explain complex concepts in one simple paragraph.
- ✓ To check understanding, use prompts that generate quiz questions on the material.
- ✓ Give examples and non-examples of specific concepts.

Generative AI excels at summarizing and has been used to make content more accessible. Review AI-generated summaries for accuracy.



Summarization and Comprehension Checks

Examples/Prompts:

- “Summarize the main ideas of this chapter in 4 bullet points.”
- “Explain this concept in one simple paragraph.”
- “What are 3 quiz questions (and answers) on this material?”
- “Give examples of [concept] and non-examples.”
- “Summarize this War of 1812 article into a timeline of 5 key events.”
- “Generate a short summary of the experiment we did and then give me two true/false questions with answers.”



Social and Behavioural Supports

What AI can do: AI can **write social stories and scripts** that model appropriate behaviour or social interactions.

Why it helps: Students with autism or social/emotional needs often learn best from concrete, structured stories that explain “what to do” in common situations. Traditionally teachers or aides write these social stories by hand. AI can speed up this process and tailor the language/characters to the student.

Social scripts **reduce anxiety and clarify expectations.**



Social and Behavioural Supports

How to use it:

- ✓ Prompt ChatGPT with a scenario and ask it to generate a social story.
- ✓ Create lists and scripts For behavior expectations in specific situations.
- ✓ Turn key sentences from social stories and scripts into picture cards.

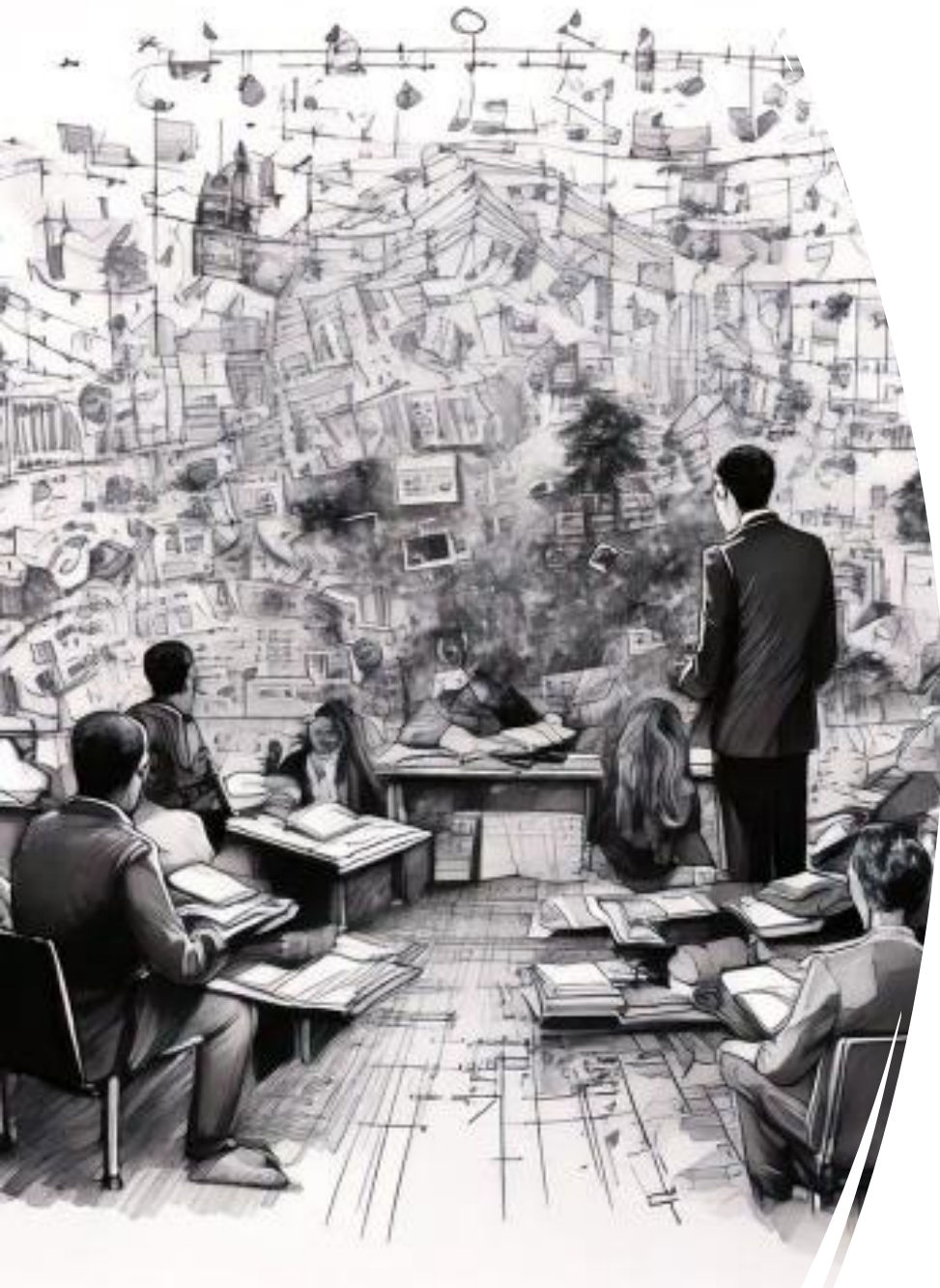
Tools like “Social Scripts Generators” have been noted as especially helpful for autistic students learning classroom expectations. Generative AI’s conversational nature allows students to rehearse social communication in a safe, non-judgmental way. Always review for appropriateness and inclusivity.



Social and Behavioural Supports

Examples/Prompts:

- “Write a social story (simple narrative) about starting the day in high school: including walking into school, going to class, and saying hello to the teacher. Use first-person and simple language.”
- “Create a list of polite phrases a student could use if someone is interrupting them.”
- “Draw a cartoon of a student raising his hand to speak”
- “Create a 4-step social story for staying calm in the hallways. Include one sentence about deep breathing.”
- “Write a friendly conversation script for a student asking a question after class.”

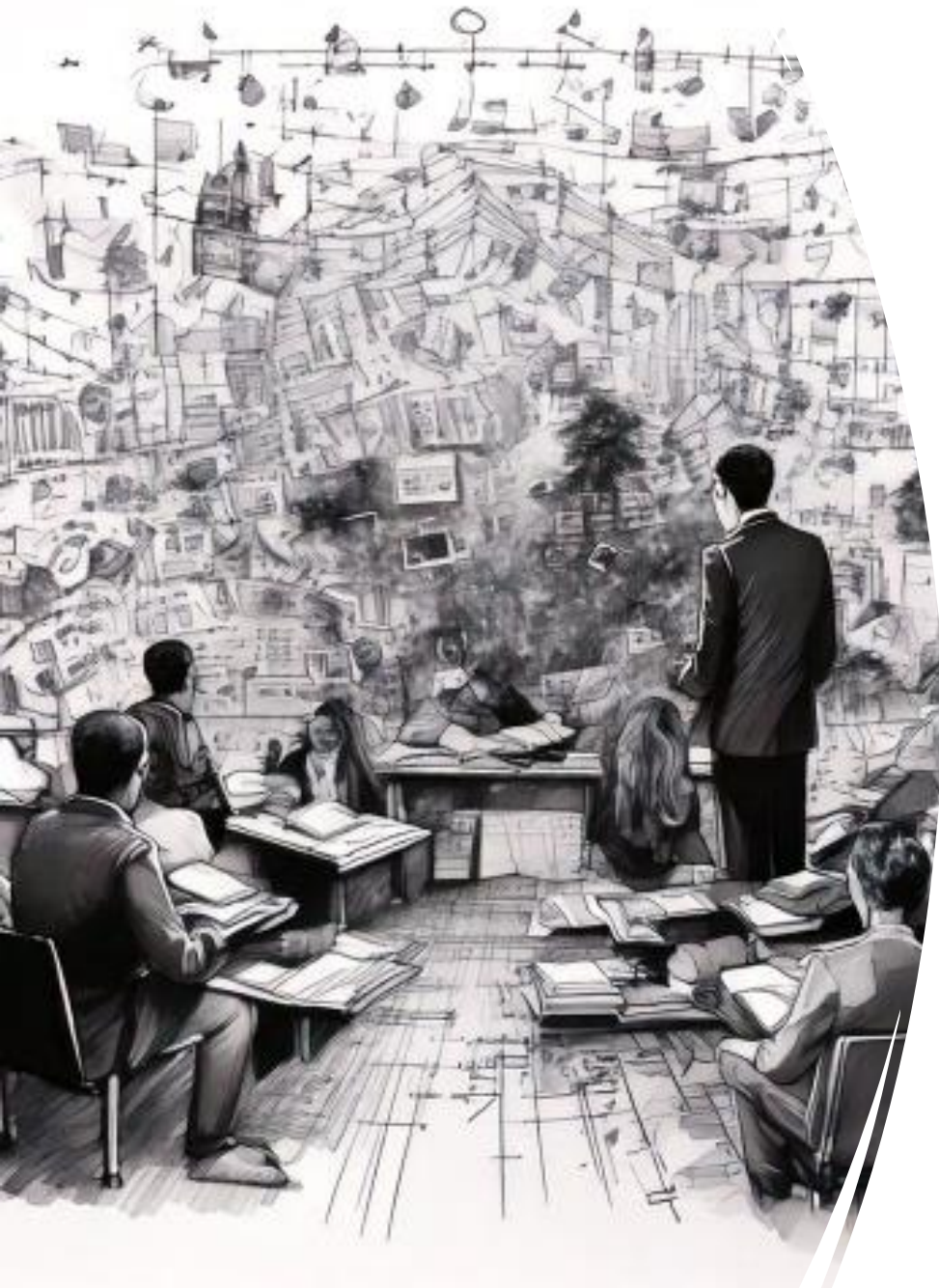


Writing and Communication Assistance

What AI can do: Generative AI can **assist with writing, spelling and communication**. For students with writing difficulties (dysgraphia, ADHD), AI can suggest better phrasing, correct grammar, or expand on a brief idea. AI can also check tone for social communication.

Why it helps: Students with dyslexia or dysgraphia often know the content but struggle to express it in writing. AI editors **provide immediate feedback** without the stigma of correction from a peer. This fosters confidence in drafting.

Multilingual learners can use AI to check their English writing or to practice writing in their home language.

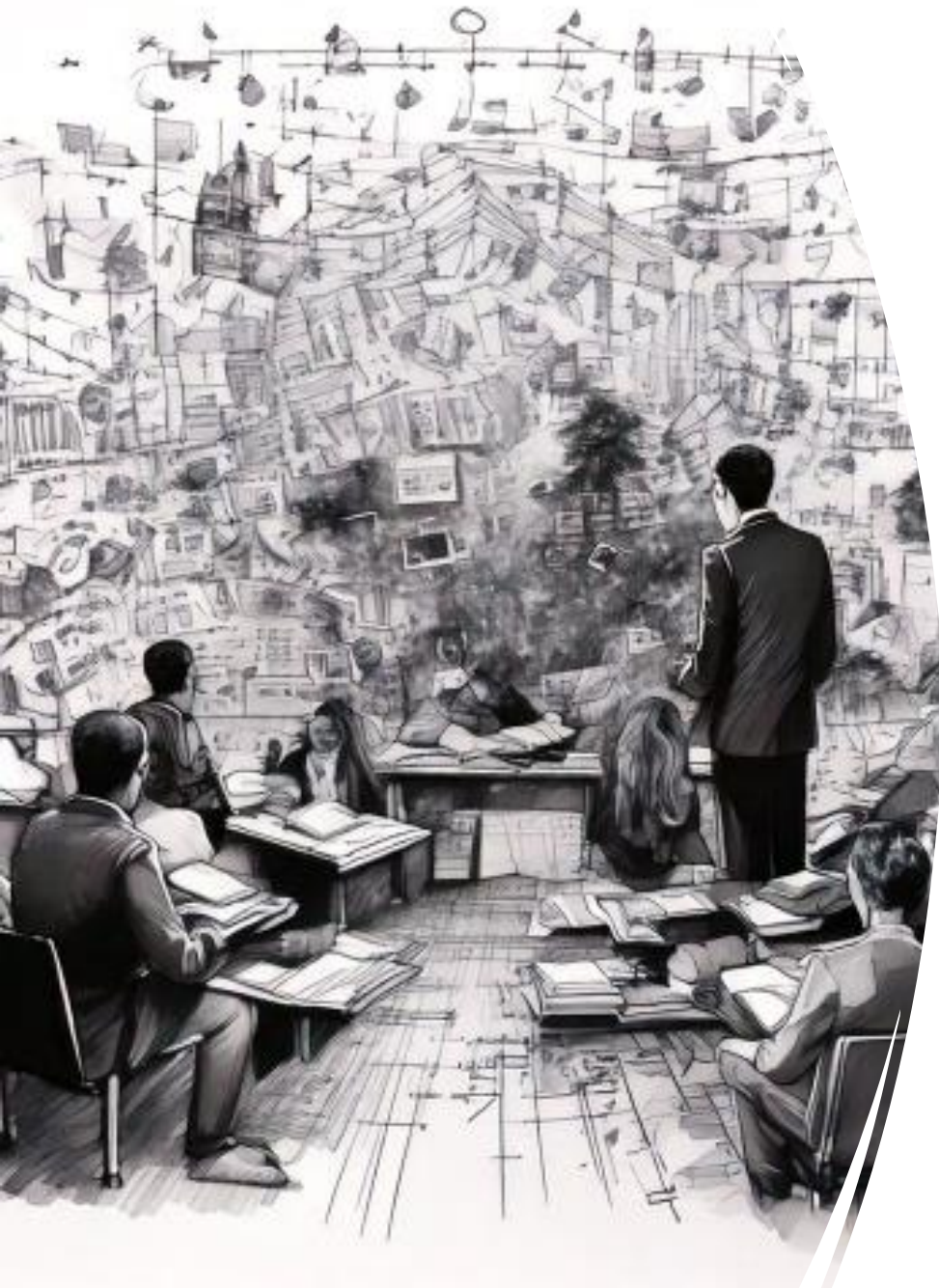


Writing and Communication Assistance

How to use it:

- ✓ Add text to ChatGPT, Copilot, or Gemini and ask the assistant to rephrase or check it.
- ✓ Chatbots can be prompted expand on ideas or to correct spelling and grammar.
- ✓ Encourage students to iterate: if the AI's suggestions aren't perfect, they can refine the prompt (e.g. "shorter sentences", "no advanced vocabulary").
- ✓ Use text-to-speech features in AI tools to hear text read aloud for additional support.

AI can proofread and refine written content, improving readability and accessibility for students with dyslexia or difficulties in writing.



Writing and Communication Assistance

Examples/Prompts:

- “My answer is: [student’s sentence]. Please expand this into two sentences that explain the idea more fully.”
- “Correct the spelling/grammar of this paragraph.”
- “I wrote this science conclusion. Please improve the clarity and fix any mistakes: [student text].”
- “Rewrite this email to my teacher so it sounds polite and clear.”



Practical Tips and Templates

- **Level Adjustments:** When simplifying text, include *guidance for reading level*. E.g. “Make this text suitable for a Grade 8 reader (age 13–14), with short sentences and common words.”
- **Step-by-Step Prompts:** For task breakdowns or procedures, use ordered lists: “List steps 1, 2, 3...” This gives structured output.
- **Check and Edit AI Output:** Always review the AI’s response. Adjust the prompt (e.g. “Simpler”, “more detail”) if needed.
- **Use Accessible Language in Prompts:** Model how to speak to AI. For example, “*Explain it to me like I’m 12 years old.*” often yields simpler language.



Practical Tips and Templates

- **Prompt Templates:** Create fill-in templates that can be reused. For instance:
 - *“Create a beginner-level worksheet about [topic], including 3 questions and answers.”*
 - *“Convert this text [paste] into a fill-in-the-blank quiz.”*



Conclusion and Best Practices

- ✓ **Supportive Scaffolding:** Generative AI can adapt and scaffold learning by simplifying text, adding visuals, structuring tasks, and creating alternative formats.
- ✓ **Practical Potential:** AI Tools can help transform generic materials into personalized resources, aligning with the broader goal of adaptive learning for students.
- ✓ **Assistive, Not Autonomous:** AI tools support but do not replace human judgment or care; always check content for accuracy and use them ethically and sensitively.
- ✓ **Privacy and Inclusion:** Respect school privacy policies—avoid sharing identifiable student information—and actively involve students in choosing the formats that support their learning best.